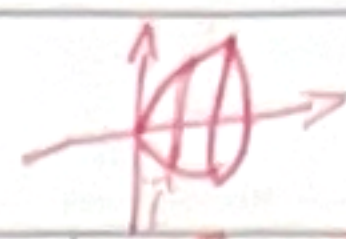


$$\int_0^1$$

围着x轴转 360°



切片是工機轉車由切刀

半径在包围区上

$(\frac{1}{2} \pi, \frac{1}{2} \pi) \rightarrow (\frac{1}{2} \pi, \frac{1}{2} \pi)$

半径在

$x=1$

$$x =$$

The volume is

$$V = \pi \int_0^1 y^2 dx = \pi \int_0^1 x dx = \pi \left(\frac{1}{2} x^2 \right) \Big|_0^1 = \frac{\pi}{2}$$

$$V = \pi \int_0^1 y^2 dx = \pi \int_0^1 x^2 dx = \left(\frac{\pi}{3} x^3 \right) \Big|_0^1 = \frac{\pi}{3}$$

$$V = \pi \int_0^1 y^2 dx = \pi \int_0^1 x dx = \left(\frac{\pi}{2} x^2 \right) \Big|_0^1 = \frac{\pi}{2}$$

The volume is

$$V = \pi \int_0^1 x^4 dx = \pi \left(\frac{1}{5} x^5 \right) \Big|_0^1 = \frac{\pi}{5}$$

The volume is

$$V = \pi \int_0^1 (x^2)^2 dx = \pi \left(\frac{1}{5} x^5 \right) \Big|_0^1 = \frac{\pi}{5}$$

The volume is

$$V = \pi \int_0^1 (x^2)^2 dx = \pi \left(\frac{1}{5} x^5 \right) \Big|_0^1 = \frac{\pi}{5}$$

* The volume is

$$V = \pi \int_0^1 (x)^2 dy = \pi \int_0^1 y dy = \frac{\pi}{2} (y^2) \Big|_0^1 = \frac{\pi}{2}$$

The volume is

$$V = \pi \int_0^1 (\sqrt{y})^2 dy = \pi \int_0^1 y dy = \frac{\pi}{2} (y^2) \Big|_0^1 = \frac{\pi}{2}$$

~~The volume is~~

$$V = \pi \int_0^1 (x^2) dx$$

The volume is

$$V = \int_0^1 \pi(\sqrt{y})^2 dy = \int_0^1 \pi y dy = \frac{1}{2} \pi (y^2) \Big|_0^1 = \frac{\pi}{2}$$

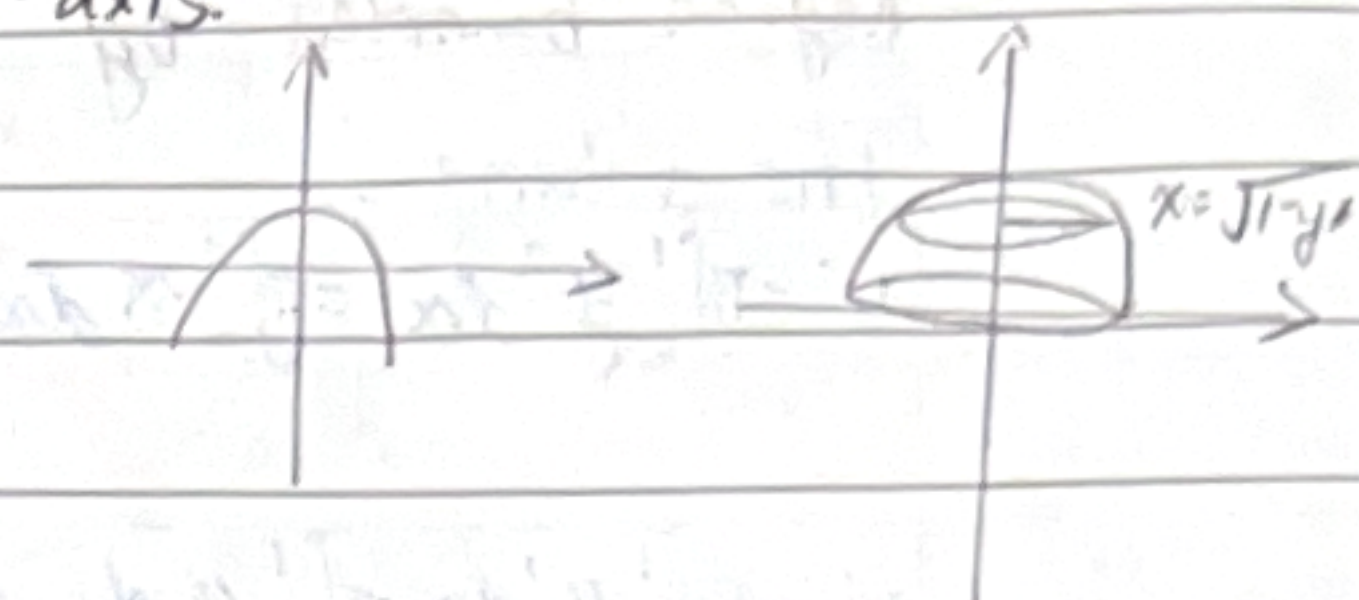
3rd week

$$x = \sqrt{1-y}$$

- 4 Region bounded by $y = 1 - x^2$ and $y = 0$ about the y -axis.

The volume is

$$V = \int_0^1 \pi x^2 dy = \pi \int_0^1 (1-y) dy = \pi \left(y - \frac{1}{2} y^2 \right) \Big|_0^1 = \pi \left(1 - \frac{1}{2} \right) = \frac{\pi}{2}$$



The volume is

$$V = \int_0^1 \pi (\sqrt{1-y})^2 dy = \pi \int_0^1 (1-y) dy = \pi \left(y - \frac{1}{2} y^2 \right) \Big|_0^1 = \pi \left(1 - \frac{1}{2} \right) = \frac{\pi}{2}$$

The volume is

$$V = \int_0^1 \pi (\sqrt{1-y})^2 dy = \pi \int_0^1 (1-y) dy = \pi \left(y - \frac{1}{2} y^2 \right) \Big|_0^1 = \pi \left(1 - \frac{1}{2} \right) = \frac{\pi}{2}$$

- 5 Region bounded by $y = x^2$ and $y = x$ about the x -axis.

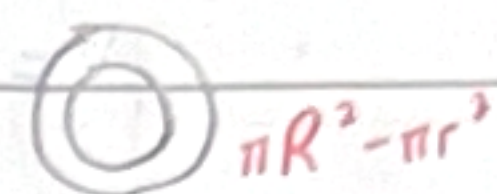
The volume is

$$V = \pi \int_0^1 (x^2 - (x^2)^2) dx = \pi \int_0^1 (x^2 - x^4) dx = \pi \left(\frac{1}{3} x^3 - \frac{1}{5} x^5 \right) \Big|_0^1 = \frac{2\pi}{15}$$



The volume is

$$V = \pi \int_0^1 (\pi x^2 - \pi (x^2)^2) dx = \pi \int_0^1 (x^2 - x^4) dx = \pi \left(\frac{1}{3} x^3 - \frac{1}{5} x^5 \right) \Big|_0^1 = \frac{2\pi}{15}$$



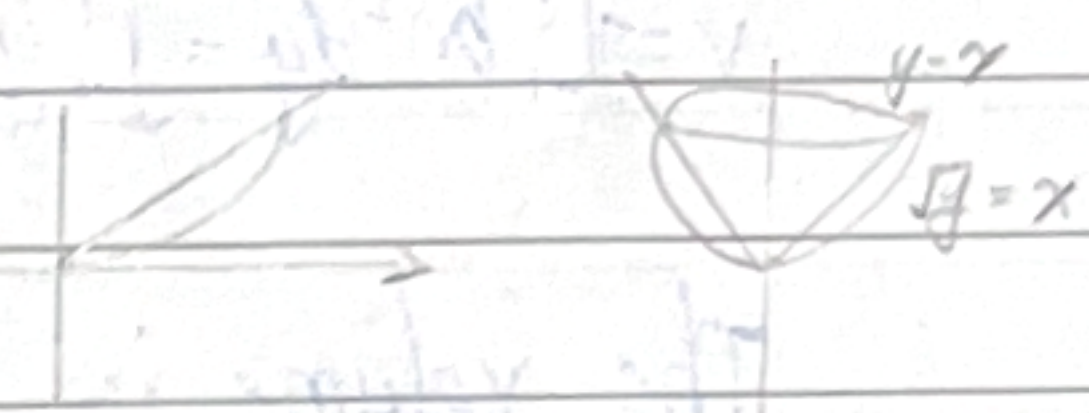
The volume is

$$V = \pi \int_0^1 (\pi (x^2)^2 - \pi (x^2)^2) dx = \pi \int_0^1 (x^2 - x^4) dx = \pi \left(\frac{1}{3} x^3 - \frac{1}{5} x^5 \right) \Big|_0^1 = \frac{2\pi}{15}$$

- 6 Region bounded by $y = x^2$ and $y = x$ about the y -axis.

The volume is

$$V = \pi \int_0^1 ((\sqrt{y})^2 - y^2) dy = \pi \int_0^1 (y - y^2) dy = \pi \left(\frac{1}{2} y^2 - \frac{1}{3} y^3 \right) \Big|_0^1 = \frac{\pi}{6}$$



The Volume is

$$V = \int_0^1 (\pi (\sqrt{y})^2 - \pi (y)^2) dy = \pi \int_0^1 (y - y^2) dy = \pi \left(\frac{1}{2} y^2 - \frac{1}{3} y^3 \right) \Big|_0^1 = \pi \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{\pi}{6}$$

The Volume is

$$V = \int_0^1 (\pi (\sqrt{y})^2 - \pi (y)^2) dy = \pi \int_0^1 (y - y^2) dy = \pi \left(\frac{1}{2} y^2 - \frac{1}{3} y^3 \right) \Big|_0^1 = \pi \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{\pi}{6}$$

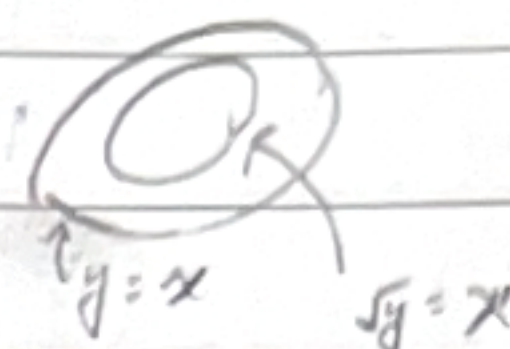
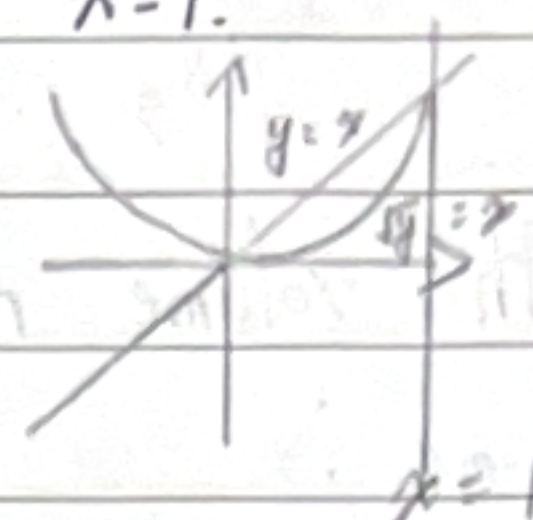
3rd week

7. Region bounded by $y=x^2$ and $y=x$ about the line $x=1$.

The Volume is

$$V = \pi \int_0^1 ((1-y)^2 - (1-\sqrt{y})^2) dy = \pi \int_0^1 (1-2y+y^2 - 1+2\sqrt{y}-y^2) dy$$

$$= \pi \int_0^1 (2\sqrt{y}-2y) dy$$



The Volume is

$$V = \int_0^1 (\pi(1-y)^2 - \pi(1-\sqrt{y})^2) dy$$

$$= \pi \int_0^1 (1-2y+y^2 - 1+2\sqrt{y}-y^2) dy = \pi \int_0^1 (-2y+y^2+2\sqrt{y}) dy$$

$$= \pi \left(-\frac{2}{2}y^2 + \frac{1}{3}y^3 + \frac{4}{3}y^{3/2} \right) \Big|_0^1 = \pi \left(-1 + \frac{1}{3} + \frac{4}{3} \right) = \frac{2\pi}{3}$$

The Volume is

$$V = \int_0^1 (\pi(1-y)^2 - \pi(1-\sqrt{y})^2) dy = \pi \int_0^1 (-2y+y^2-y+2\sqrt{y}) dy$$

$$= \pi \int_0^1 (y^2-3y+2\sqrt{y}) dy = \pi \left(\frac{1}{3}y^3 - \frac{3}{2}y^2 + \frac{4}{3}y^{3/2} \right) \Big|_0^1 = \pi \left(\frac{1}{3} - \frac{3}{2} + \frac{4}{3} \right) = \frac{\pi}{6}$$

The Volume is

$$V = \int_0^1 (\pi(1-y)^2 - \pi(1-\sqrt{y})^2) dy = \pi \int_0^1 (-2y+y^2-y+2\sqrt{y}) dy$$

$$= \pi \int_0^1 (y^2-3y+2\sqrt{y}) dy = \pi \left(\frac{1}{3}y^3 - \frac{3}{2}y^2 + \frac{4}{3}y^{3/2} \right) \Big|_0^1 = \pi \left(\frac{1}{3} - \frac{3}{2} + \frac{4}{3} \right) = \frac{\pi}{6}$$

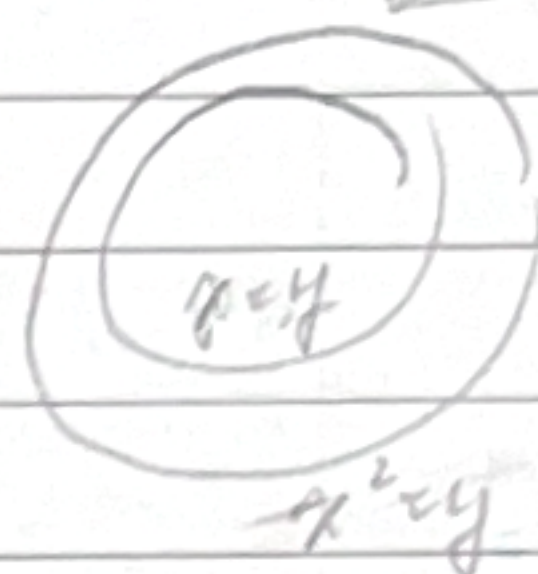
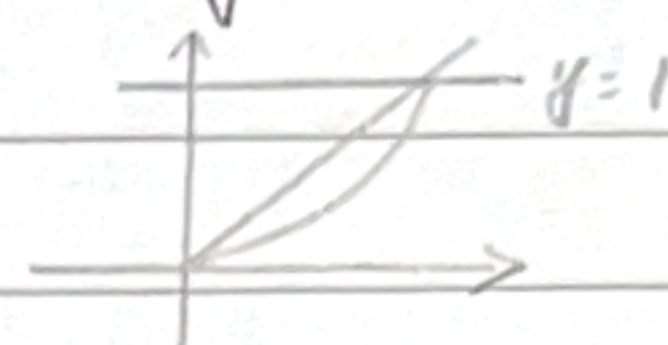
8. Region bounded by $y=x^2$ and $y=x$ about the line $y=1$.

The Volume is

$$V = \int_0^1 (\pi(1-x^2)^2 - \pi(1-x)^2) dx = \pi \int_0^1 (-2x^2+x^4-x^2+2x) dx$$

$$= \pi \int_0^1 (x^4-3x^2+2x) dx = \pi \left(\frac{1}{5}x^5 - x^3 + x^2 \right) \Big|_0^1$$

$$= \pi \left(\frac{1}{5} - 1 + 1 \right) = \frac{\pi}{5}$$



The Volume is

$$V = \int_0^1 (\pi(1-x^2)^2 - \pi(1-x)^2) dx = \pi \int_0^1 (-2x^2+x^4-x^2+2x) dx$$

$$= \pi \int_0^1 (x^4-3x^2+2x) dx = \pi \left(\frac{1}{5}x^5 - x^3 + x^2 \right) \Big|_0^1$$

$$= \pi \left(\frac{1}{5} - 1 + 1 \right) = \frac{\pi}{5}$$

The Volume is

$$V = \int_0^1 (\pi(1-x^2)^2 - \pi(1-x)^2) dx = \pi \int_0^1 (-2x^2+x^4-x^2+2x) dx$$

$$= \pi \int_0^1 (x^4-3x^2+2x) dx = \pi \left(\frac{1}{5}x^5 - x^3 + x^2 \right) \Big|_0^1$$

$$= \pi \left(\frac{1}{5} - 1 + 1 \right) = \frac{\pi}{5}$$

3rd week

Cylindrical Shell Method

通过圆周长寻找体积 $\approx 2\pi r h \Delta x$

$$[2\pi rh \Delta x] V = \int_a^b 2\pi (\text{radius}) (\text{height}) dx$$

rotation of y-axis
rotation of x-axis

$$V = \int_a^b 2\pi (\text{radius of } x) (h(x)) dx$$

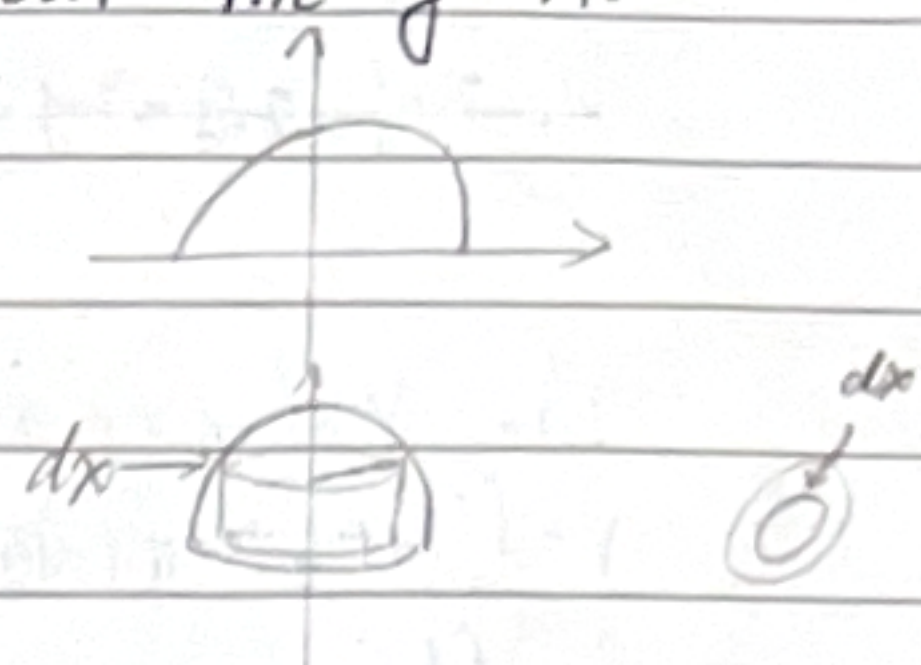
$$V = \int_c^d 2\pi (\text{radius of } y) (h(y)) dy$$

Find the volume region bounded by $y=1-x^2$ and $y=0$ about the y-axis

The Volume is

$$V = \int_0^1 2\pi x(1-x^2) dx = 2\pi \int_0^1 (x-x^3) dx$$

$$= 2\pi \left(\frac{1}{2}x^2 - \frac{1}{4}x^4 \right) \Big|_0^1 = 2\pi \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{\pi}{2}$$



The Volume is

$$V = 2 \int_0^1 2\pi x(1-x^2) dx = 4\pi \int_0^1 (x-x^3) dx$$

$$= 4\pi \left(\frac{1}{2}x^2 - \frac{1}{4}x^4 \right) \Big|_0^1 = 4\pi \left(\frac{1}{2} - \frac{1}{4} \right) = \pi$$

The Volume is

$$V = 2 \int_0^1 2\pi x(1-x^2) dx = 4\pi \int_0^1 (x-x^3) dx$$

$$= 4\pi \left(\frac{1}{2}x^2 - \frac{1}{4}x^4 \right) \Big|_0^1 = 4\pi \left(\frac{1}{2} - \frac{1}{4} \right) = \pi$$